# Formulae & Definitions — Topic 17 · Oscillations

Period 周期 T — the time for one complete oscillation: $T = 2\pi/\omega$ (s).

Frequency 频率 f — the number of oscillations per unit time: $f = 1/T$ (Hz).

Angular frequency 角频率 ω — $\omega = 2\pi/T = 2\pi f$ (rad $s^{-1}$).

Displacement 位移 x — the distance from the equilibrium position (m).

Amplitude 振幅 $x_0$ — the maximum displacement from the equilibrium position (m).

Phase 相位 / Phase difference 相位差 — the point reached in a complete cycle; phase difference can be given as a fraction of a cycle, in degrees or radians.

Simple harmonic motion 简谐振动 — an oscillation in which the acceleration is always proportional to the displacement and directed towards the equilibrium position: $a = -\omega^2 x$ (defining equation). ω is constant and independent of amplitude.

SHM displacement & velocity — if $x = x_0 \sin(\omega t)$ (starting at equilibrium) then $v = v_0 \cos(\omega t)$; if $x = x_0 \cos(\omega t)$ (starting at an extreme) then $v = -v_0 \sin(\omega t)$; $v_0 = \omega x_0$.

v in terms of x — $v = \pm\omega\sqrt{(x_0^2 - x^2)}$; $v = v_0$ at $x = 0$, $v = 0$ at $x = x_0$.

Energy in SHM — total energy = KE + PE = $\frac{1}{2}m\omega^2 x_0^2$ (constant); PE = $\frac{1}{2}m\omega^2 x^2$, KE = $\frac{1}{2}m\omega^2(x_0^2 - x^2)$. Energy swaps between PE and KE every half-cycle.

Damping 阻尼 — loss of energy to the surroundings; light / heavy / critical damping (critical returns to equilibrium without oscillating).

Free oscillation 自由振动 — oscillating at its own natural frequency (independent of amplitude).

Forced oscillation 受迫振动 — oscillation driven by an external periodic force; it oscillates at the driving frequency.

Resonance 共振 — maximum amplitude when the driving frequency equals the natural frequency.

Effect of damping on resonance — damping lowers the resonant frequency, reduces the peak amplitude and broadens the peak.

Uses of resonance — musical instruments, microwave cooking, MRI; damaging example: earthquakes (dampers reduce the amplitude).

点评 · Insight

重点是 S.H.M 的定义：用牛顿运动定律分析振动系统，得到 a 与 x 的关系满足 $a = -\omega^2 x$
即证明是简谐振动（否则不是）。周期/频率是固定常量（natural frequency），与振幅无关。